
HS 548: Science and Society

Course Code	HS 548
Course Name	Science and Society
L-T-P-C	3-0-0-3
Prerequisites	None
Intended for	Ph.D., Masters, Advanced B.Tech. students
Mutual Exclusion	None
Approval	50th BoA (Jul 2023)

Course Contents

- Unit 1: Science, Values and the Social: (12 hours)
 - What is Science?
 - The ‘Epistemic’, the ‘Cognitive’, and the ‘Social’ in Science
 - Is Science Value-Free?
 - The Normative Structure of Science
 - Normal Science and the Structure of Scientific Revolutions
 - Scientific Objectivity
- Unit 2: Construction of Facts (10 hours)
 - Construction of Scientific Facts
 - ‘Social’ in ‘Science’ and ‘Science’ in ‘Social’ or the Mutual Constitution of Science and Society
 - Scientific Objects
 - Truth and Representation in the Sciences
 - Circulating Reference: ‘Context’ and ‘Content’ of Science
- Unit 3 Practices of Science (10 hours)
 - Scientific Practices
 - Epistemic Cultures
 - Science as a Vocation, Technical Life, and the Scientist as an Individual
 - Social Epistemology of Experiments
 - (En)gendered Science
- Unit 4 Science, Democracy and Governance (10 hours)
 - Expertise and Science
 - Politics of Science: Science as Ideology
 - Infrastructural Imaginaries
 - Debating Science Policies
 - Science and Citizens

Text Books

1. Machamer, P and Wolters, G (Eds.). 2004. Science, Values, and Objectivity. Pittsburgh: University of Pittsburgh.
2. Sismondo, Sergio. 2004. An Introduction to Science and Technology Studies. Malden, USA. Blackwell Publishing Ltd.

References

1. Biagioli, M (Ed.). 1999. The Science Studies Reader. New York, NY: Routledge.
2. Daston, L (Ed.). 2000. Biographies of Scientific Objects. Chicago: University of Chicago Press.
3. Daston L, Galison P. 1992. 'The Image of Objectivity'. Representations 40 (special issue): 81-128.
4. Douglas, H. 2009. Science, Policy, and the Value-free Ideal. Pittsburgh: University of Pittsburgh.
5. Fleck, L. 1979. Genesis and Development of a Scientific Fact. Chicago: The University of Chicago Press.
6. Galison, P. 1987. How Experiments End. Chicago: The University of Chicago Press.
7. Golinski, Jan. 1998. Making Natural Knowledge: Constructivism and the History of Science. Cambridge, UK: Cambridge University Press.
8. Hess, D. 1997. Science Studies: An Advanced Introduction. New York: New York University Press.
9. Jasanoff, S (Ed.). 2004. States of Knowledge: The Co-Production of Science and Social Order. London: Routledge.
10. Kuhn, T.S. 2012. The Structure of Scientific Revolution. Chicago: University of Chicago Press.
11. Knorr Cetina, K. 1999. Epistemic Cultures: How the Sciences Make Knowledge. Cambridge, Massachusetts: Harvard university Press.
12. Latour, B. and Woolgar, S. 1986. Laboratory Life: The Construction of Scientific Facts. Princeton, New Jersey. Princeton University Press.
13. Latour, B. 1999. Pandora's Hope: Essays on the Reality of Science Studies. Cambridge: Harvard University Press.
14. Leach, M, I. Scoones, and B. Wynne (Eds.). 2005. Science and Citizens: Globalization and the Challenge of Engagement. London: Zed Books.
15. Lewontin, R. 1992. Biology as Ideology: The Doctrine of DNA. New York: Harper Perennial.
16. Merton, R, N. Storer. 1973. The Sociology of Science: Theoretical and Empirical Investigations. Chicago: The University of Chicago Press.
17. Mitchell, T. 2002. Rule of Experts: Egypt, Techno-politics, Modernity. Berkeley: University of California Press.
18. Porter, T.M. 2020. Trust in Numbers: The Pursuit of Objectivity in Science and Public Life. Princeton, New Jersey: Princeton University Press.
19. Sarukkai, S. 2005. 'Revisiting the 'Unreasonable Effectiveness' of Mathematics'. Current Science, Vol. 88(3): 415-423.
20. Sarukkai, S. 2012. What is Science? New Delhi: National Book Trust, India.
21. Shapin, S and Schaffer, S. 1985. Leviathan and the Air-Pump: Hobbes, Boyle, and the Experimental Life. Princeton: Princeton University Press.
22. Sukumar, AM. 2019. Midnight's Machines: A Political History of Technology in India. New York: Penguin Random House.
23. Weber, M. 1946. 'Science as a Vocation'. In Gerth HH and C Wright Mills (Eds.). From Max Weber: Essays in Sociology. New York: Oxford University Press, 129-156.